

AI & Automation Final Test -2

Instructions:

- 1) All questions are compulsory.
- 2) Each solution must be presented in a runnable format using packages where necessary, with sufficient setup/import statements for it to run.
- 3) Each question has a given time limit, after which a submission will not be accepted.
- 4) Finishing a question early does not add the leftover time to the following question.
- 5) Internet access and documentation are allowed, but not ChatGPT-generated solutions or previously completed team/course solutions.
- 6) Partial marks will be awarded where necessary.
- 7) All submissions must be pushed to GitHub before the time limit, failure to do so will result in the submission being invalid.
- 8) The bonus question is independent of the previous questions and is not considered for the total marking of the paper. Marks obtained on this question are incremented to your score without affecting the total possible score of the paper.
- 9) All questions are 10 points, and the bonus question is 5 points.

Questions:

1. A mobile robot publishes its yaw on `/imu/yaw` (`std_msgs/Float32`), its position on `/gps/x` and `/gps/y` (`std_msgs/Float32`), and accepts velocity commands on `/cmd_vel` (`geometry_msgs/Twist`). Write a class-based ROS 2 C++ node that subscribes to the three sensor topics, maintains the robot's current (x, y) position and yaw, and accepts a target (x, y) through ROS parameters. The node should compute the heading error to the target and publish a Twist command: rotate when the heading error is large, move forward when aligned, and stop when within 0.5 m of the target. Use appropriate publishers, subscribers, callbacks, and parameters. (12 mins)
2. A video contains a red object moving against a relatively uniform background. Using OpenCV, write a program that converts each frame to an appropriate colour space, segments the red object, removes noise using morphological operations, and finds contours to select the largest valid one. Draw its bounding box and centroid on each frame, and estimate the object's movement direction using its centroid in consecutive frames. Do not use a pretrained object-detection model. (10 mins)
3. You are given a CSV containing the fields age, temperature, speed, distance, battery, failed. Write a program that loads the dataset and handles missing values, splits the data into training and testing sets, and standardizes the numerical features. Train a Logistic Regression model and a Decision Tree model, and evaluate both using Accuracy, Precision, Recall, and F1-score. Perform 5-fold cross-validation and determine which model performs better. Also explain why accuracy alone may be misleading when the classes are imbalanced. (12 mins)
4. You are given a dataset containing images of three rover components, organised in the folders `switch/`, `knob/`, and `socket/`. Write a program that loads the dataset and creates training/validation splits, resizes and normalizes the images, and applies suitable data augmentation. Build a

CNN-based classifier, select an appropriate loss function and optimizer, train the model, and report the validation accuracy. Finally, run inference on a new image. A pretrained CNN and transfer learning may be used. (12 mins)

5. Write a C++ program that receives a `std::vector<float>` ranges of LiDAR distance measurements. Remove invalid values (NaN, infinity, and negative values), then calculate the minimum valid distance and the average valid distance. Count the measurements below an obstacle threshold of 1.0 m, and return the index of the closest valid measurement. Use functions, STL containers, and appropriate C++ constructs, and do not use external libraries. The program must output: Minimum distance, Average distance, Obstacles detected, and Closest measurement index. (10 mins)

Bonus Question:

1. Create a ROS 2 launch file that starts both `perception_node` and `navigation_node`. The launch file should accept two launch arguments named exactly `camera_topic` (default: `/camera/image_raw`) and `lidar_topic` (default: `/scan`), and pass them as ROS parameters to both nodes. (5 mins)